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Abstract

This research aims to uncover the relationship between race and gender, and perceptions of a person in a professional position. Respondents were shown news articles detailing a chairperson's plan to help prepare a United States community for incoming floods and heatwaves. One of four articles was randomly assigned to respondents, each including either a white chairwoman, a black chairwoman, a white chairman, or a black chairman. Dependent variables—perceived trust, competence, and ability to solve the issue—were tested on a 5-point scale. In my analysis, I found significance in dependent variables when interacting gender of the participant with their assigned article, and when interacting partisanship with the assigned article. Results showed that women were significantly more critical of white women than men were. Also, Republicans perceived black and white women as significantly more incompetent than Democrats, and perceived white women as significantly more untrustworthy and unable than Democrats.

Introduction

Diversity, Equity, and Inclusion programs in the United States have made recent headlines as a polarizing topic. The Trump administration began immediate rollbacks of DEI programs upon the start of the presidential term. This move came along with attacks against the programs for allowing under-qualified people to take professional positions. Consequently, after various tragedies, the Trump administration criticized certain professionals involved by insinuating that their race or gender helped them to get hired to a position they were not qualified

for. Specifically, after the fires that ravaged Los Angeles, Republicans accused the mayor, a black woman, and the fire chief, a white woman, for mishandling the tragedy due to their race and gender. This leads to my research question: how do race and gender affect the perceptions of a person in a professional position?

Literature Review

8 different studies on discrimination helped inform this research. The first study highlights the contextual importance of my study. Researchers used data from firms which had affirmative action policies to test if there was a significant difference in employees qualifications by race and gender. Importantly, they found that there was not a major difference, showing that inclusive policy does not affect qualifications (Holzer et al. 1996).

Two other studies focused on racial discrimination. A field experiment found that black-sounding names of job applicants received significantly less callbacks than applicants who had white-sounding names (Bertrand et al. 2004). Another study focused on applications using a labor market simulation, and found that people discriminated against candidates based on race (Wozniak et al. 2020).

Other research focused on gender discrimination against people in professional positions. This study used videos of a professional expressing frustration, and found that people believed women to be less competent in these situations (Brescoll et al. 2008). Additionally, in another study on hiring discrimination, researchers found that there is in-group favoritism for gender (Bendick et al. 2012). However, gender bias and discrimination is also perpetuated by women who are forced to use sexist structures to survive within them (Miller 2004).

One observational study looked at race and gender simultaneously. Researchers used performance evaluations to test whether race and gender affected a professional's perceived

abilities, or lead them to be judged more harshly (Bohnet et al. 2021). They found that black women were most harshly judged, regardless of qualifications.

Although there have not been many studies on differences in race or gender discrimination by political party, polling shows clear evidence that Democrats are more in favor of doing work towards racial equality (Pew Research Center 2021).

Hypotheses

I hypothesized three different relationships. First, that the black chairperson will have the lowest rating of trust, competence, and ability to solve the issue. This hypothesis is supported by Bertrand et al. 2004, and Wozniak et al. 2020 which found racial discrimination in the labor market. The second hypothesis is that if a person is the same gender as the chairperson, they will have higher ratings of trust, competence, and ability to solve the issue. This is supported by Bednick et al. 2012 which found in-group favoritism for gender. Finally, if a person identifies with a political party, their opinions of the chairperson will differ.

Experimental Design

Independent Variables

This experiment is a 2x2 factorial design. Respondents were randomly assigned one of four news stories which I created. Each story warned about incoming floods and heatwaves on Long Island, New York, but differed in one respect. Race and gender, the independent variables, of the chairperson of the Long Island Surface Water Group was different in each story. The chairperson was either a white woman, black woman, white man, or black man who was tasked with solving the issue. This treatment was made salient through the name of the chairperson, and a picture of them.

Dependent Variables

The effects of race and gender were tested through 3 dependent variables: perceived trust, competence, and ability to solve the issue. The questions asked were “how much do you trust the chairperson?”, “how competent do you believe the chairperson is?”, and “how likely is it that the chairperson will solve the issue?” Respondents were given a 5-point scale to answer each question. Low values correspond to high trust, competence, and ability, while high values correspond to high distrust, incompetence, and inability.

External Validity

Generally, field experiments have high external validity because they reproduce real life scenarios. For my experiment, I altered a pre-existing article about a real issue, and changed certain aspects to make it more applicable to my research. Therefore, my experiment has high external validity because it can be applied to other real life situations.

Sample

The sample was acquired through Lucid Theorem, to which we paid \$1.50 for each response. There were 310 total responses. I analyzed the demographics of the sample to better understand the makeup of the respondents. Ethnicity was mostly made up of white people at 234 white respondents and 73 people of color. Gender was more evenly distributed with 151 men and 159 women. Average age of the respondents was 47.8. Most participants were 54. Partisanship was also relatively evenly distributed with 113 democrats, 139 republicans, and 58 independents. Most of the respondents had just high school diplomas at 74 respondents, and the second highest group was Bachelor's degree at 65.

Balance Tests

Balance tests were conducted for gender, race, and partisanship. The p-value of gender was .876, race was .898, and partisanship was .699. This means there was not a significant difference in gender, race, or partisanship between treatment groups.

Manipulation Check

I would use manipulation check questions to ensure that treatment was effective. They would be “what is the race of the chairperson?” and “what is the gender of the chairperson?” These would be included post-survey so as to not influence these answers. A manipulation check is especially helpful to understand if a lack of significant results was because the hypothesis was rejected, or because treatment was not received.

Debrief Statement

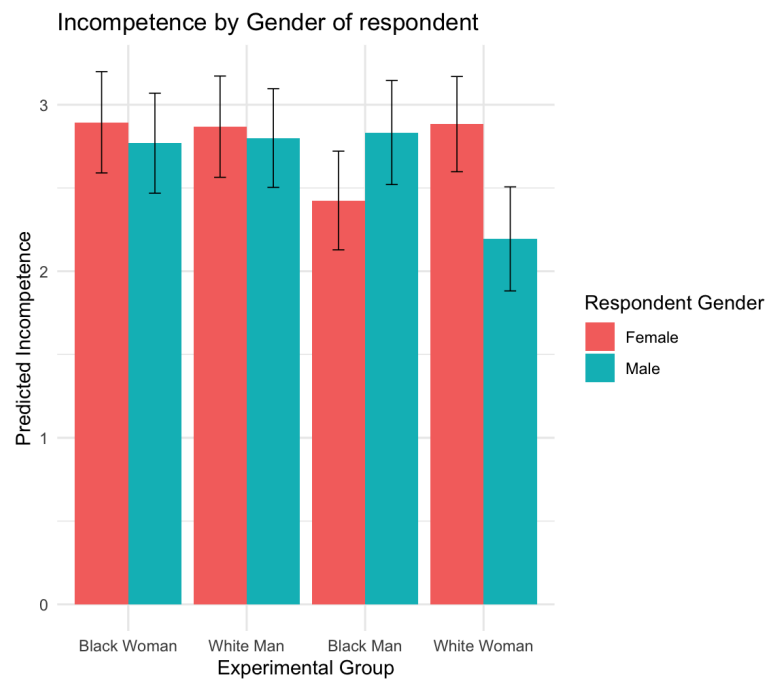
There is deception in this study. The article warns of possible tragedies to a United States community which are not real. Deception was necessary in order to make the article relevant to American citizens, fit the time constraint of the study, and effectively distribute the treatment.

Results

The first analysis was of the mean response for each dependent variable, without interacting any respondent demographic variables. Differences in means by treatment group were insignificant for each dependent variable. When interacting the treatment group with the gender of the respondent, I found significance in the difference in means for the white woman treatment for all 3 dependent variables. Race was not a significant interacting factor. Finally, I interacted treatment group with partisanship of the respondent, and found significance for trust and ability to solve the issue in the white woman, and significance for both white and black woman treatment groups for competence.

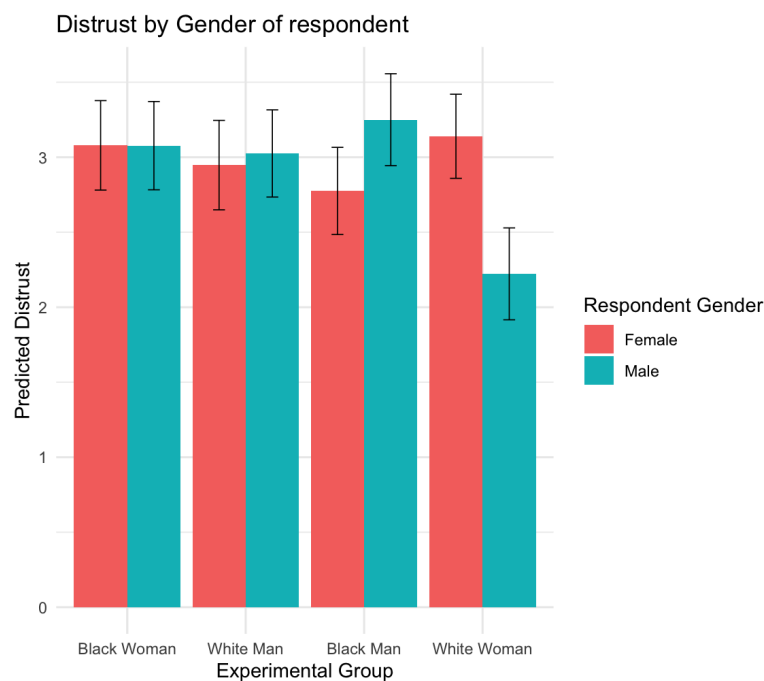
Gender

Figure 1: Incompetence by Gender of Respondent



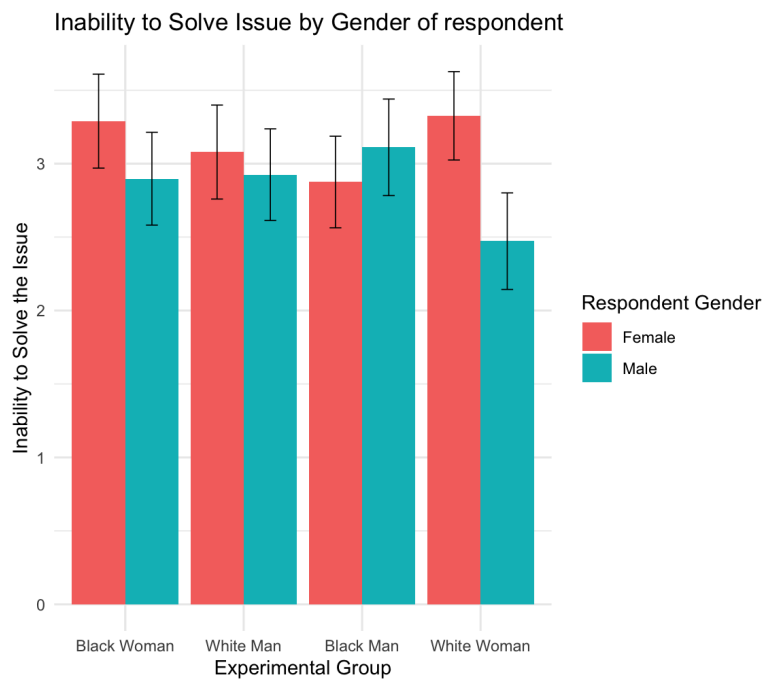
The mean for incompetence of white women when the respondent was a woman was 2.88 with confidence intervals of 2.60 and 3.17. When the respondent was a man, the mean was 2.19, with confidence intervals of 1.88 and 2.51.

Figure 2: Distrust by Gender of Respondent



For distrust, male respondents had a mean response of 3.14 in the white woman treatment, with confidence intervals of 2.86 and 3.42. Female respondents had a mean response of 2.22 with confidence intervals of 1.92 and 2.53.

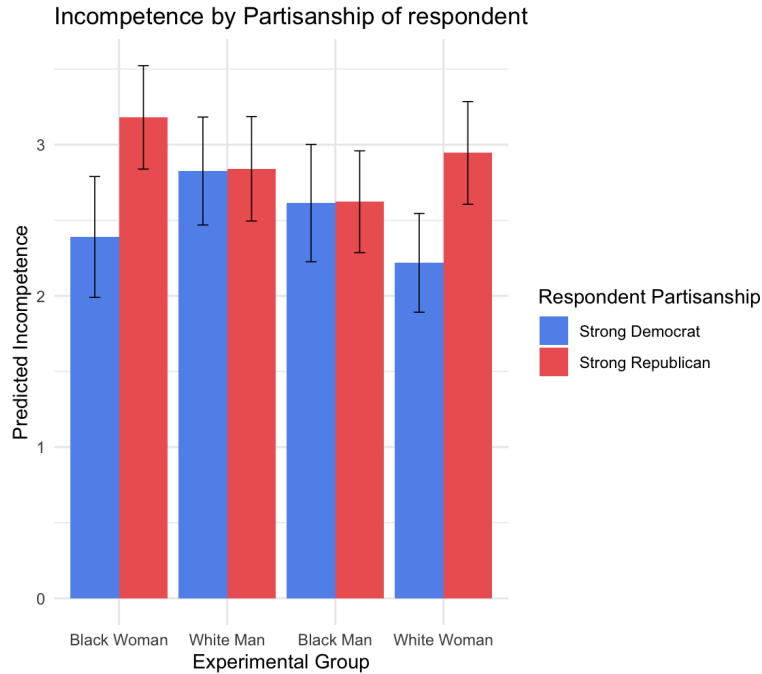
Figure 3: Inability to Solve Issue by Gender of Respondent



For the white chairwoman's inability to solve the issue, male respondents' had a mean response of 2.47, with confidence intervals of 2.14 and 2.80. Female respondents had a mean response of 3.33, with confidence intervals of 3.02 and 3.63.

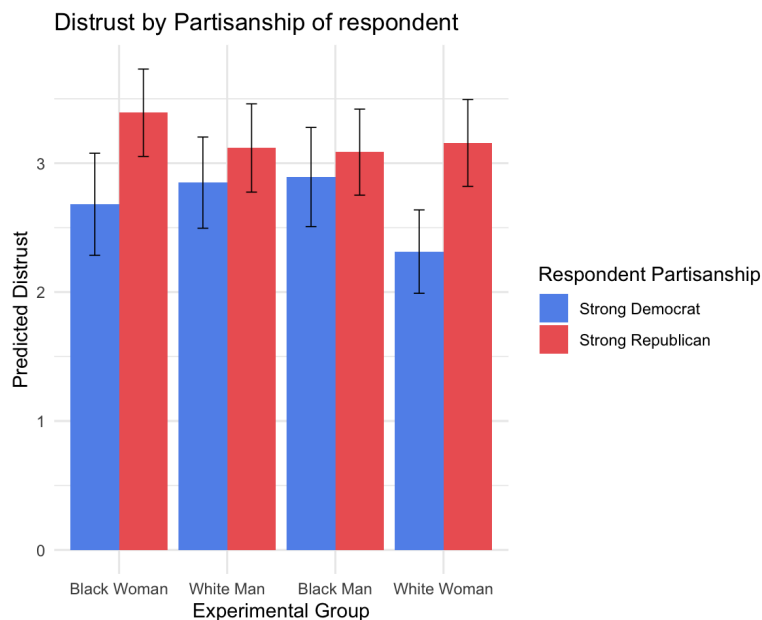
Partisanship

Figure 4: Incompetence by Partisanship of Respondent



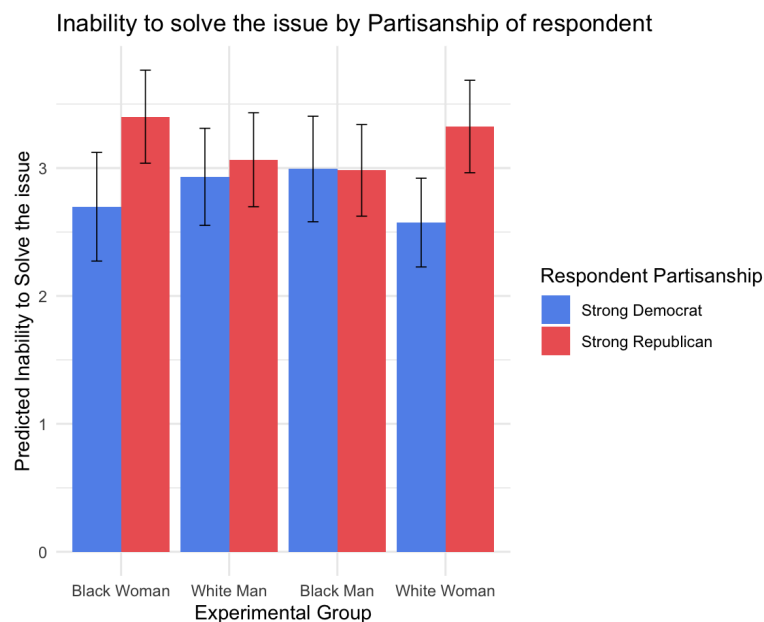
For incompetence, both the black woman treatment and white woman treatment were significant. In the white woman treatment, Republicans had a mean incompetence response of 2.95, with confidence intervals of 2.61 and 3.29. Democrats had a mean response of 2.22, with confidence intervals of 1.89 and 2.54. In the black woman treatment, Republicans had a mean response of 3.18, with confidence intervals of 2.84 and 3.52. Democrats had a mean response of 2.39, with confidence intervals of 1.99 and 2.79.

Figure 5: Distrust by Partisanship of Respondent



For distrust, only the white woman treatment was significant. In the white woman treatment, Republicans had a mean distrust response of 3.16, with confidence intervals of 2.82 and 3.49. Democrats had a mean response of 2.31, with confidence intervals of 1.99 and 2.64.

Figure 6: Inability to Solve the Issue by Partisanship of Respondent



Only the white woman treatment was significant for this dependent variable. In the white woman treatment, Republicans had a mean incompetence response of 3.32, with confidence intervals of 2.96 and 3.69. Democrats had a mean response of 2.57, with confidence intervals of 2.23 and 2.92.

Interpretation

To reiterate, low values on the 5-point response scale indicate high trust, competence, and ability, while high values indicate high distrust, incompetence, and inability. My first hypothesis

was proven untrue. There was not a significant difference in perceptions of the chairperson based on race. This is contrary to research that has proven racial discrimination. Since these studies mainly looked at the hiring process, it is possible that once a person is hired, they are no longer under the same scrutiny perpetuated by racism.

Figure 1, 2, and 3 show that female respondents perceived the white chairwoman as significantly more incompetent, untrustworthy, and unable to solve the issue than male respondents did. Although this is not what I expected to find based on my first hypothesis, it is supported by other data that suggests women uphold sexist structures (Miller 2004).

Figure 4 shows that there is a significant relationship between partisanship and perception of both the black woman and white woman. Republicans are significantly more likely to perceive both black and white women as incompetent than Democrats are. Figures 5 and 6 show significance only for the white woman treatment group. Republicans were significantly more likely to perceive the white woman as untrustworthy and unable to solve the issue than Democrats were. These findings demonstrate the relevance of partisanship divide regarding the importance of programs which move toward racial inequality as analyzed by Pew Research polling. Major topics in current political discourse clearly manifest in real life through discrimination.

Conclusion

My research has shown that recent political discourse surrounding DEI programs has made race and gender discrimination more normalized. Criticism of DEI has resulted in concerns of the qualifications of black and women professionals. This rhetoric has been led by Republicans in the Trump administration, and my research shows that other Republicans have

adopted this same rhetoric of concern. While a person's own race does not seem to play into their opinions of others, gender and partisanship have clear effects. Additionally, while research shows clear racial discrimination in the hiring process, it does not seem to be as prevalent once a person holds a professional position.

With a larger sample size, more analysis could be done. Most importantly, I would analyze the effects that intersection of a participant's race and gender have on their responses. Further experimentation can also be done to analyze the relevance that this research has in other settings. For example, there may be interesting results for professionals in different fields of labor such as blue collar workers or politicians.

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